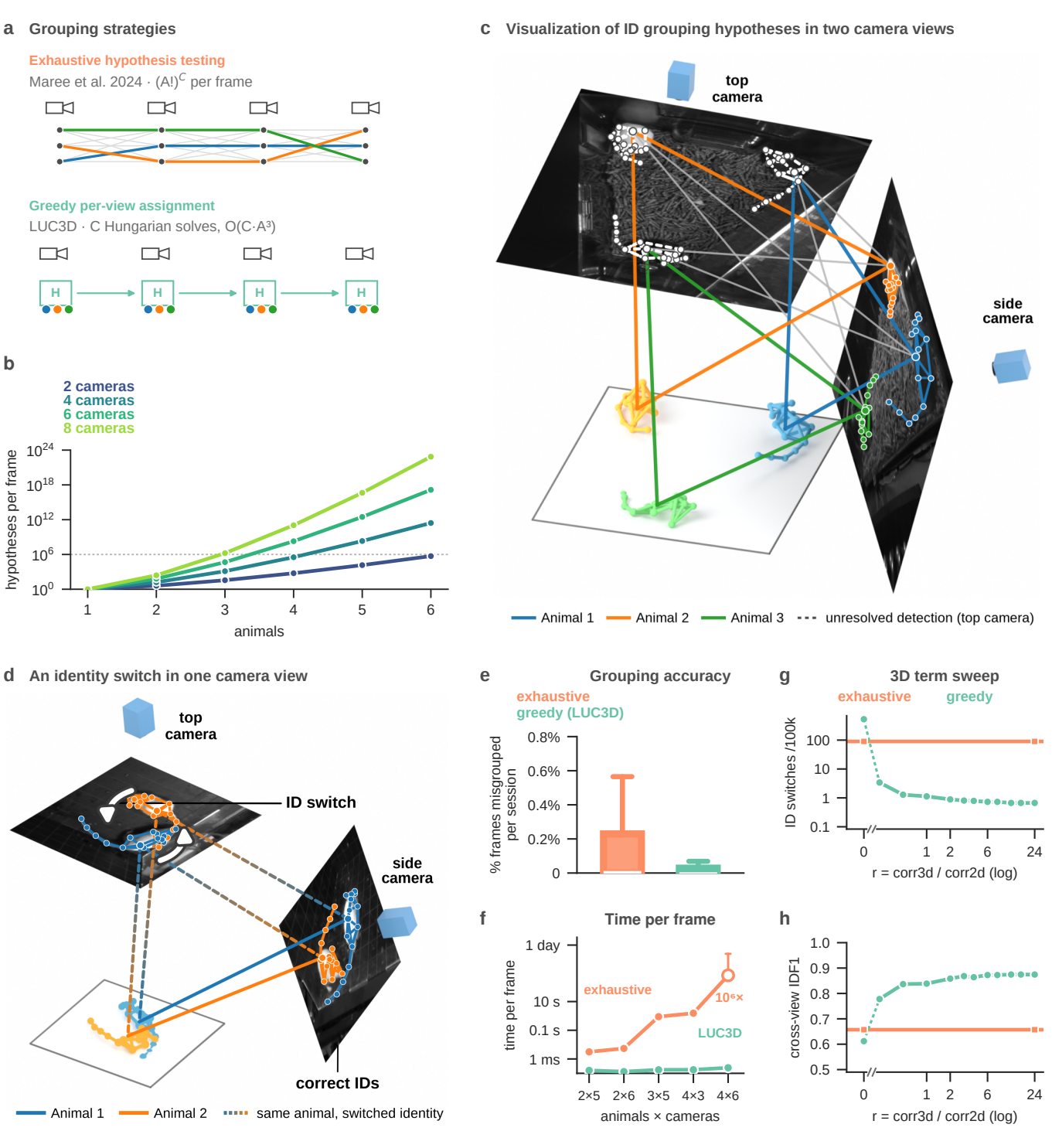

a
Grouping strategies
Exhaustive hypothesis testing
Maree et al. 2024 · (A!)^C per frame
Greedy per-view assignment
LUC3D · C Hungarian solves, O(C·A³)
H
b
2 cameras
4 cameras
6 cameras
8 cameras
hypotheses per frame
10^24
10^18
10^12
10^6
10^0
1
2
3
4
5
6
animals
c
Visualization of ID grouping hypotheses in two camera views
top camera
side camera
Animal 1
Animal 2
Animal 3
unresolved detection (top camera)
d
An identity switch in one camera view
top camera
ID switch
side camera
correct IDs
Animal 1
Animal 2
same animal, switched identity
e
Grouping accuracy
exhaustive
greedy (LUC3D)
% frames misgrouped per session
0.8%
0.6%
0.4%
0.2%
0
f
Time per frame
time per frame
1 day
10 s
0.1 s
1 ms
exhaustive
10⁶×
LUC3D
2×5
2×6
3×5
4×3
4×6
animals × cameras
g
3D term sweep
exhaustive
greedy
ID switches /100k
100
10
1
0.1
0
1
2
6
24
r = corr3d / corr2d (log)
h
cross-view IDF1
1.0
0.9
0.8
0.7
0.6
0.5
0
1
2
6
24
r = corr3d / corr2d (log)